Protecting Wildlife at Scale

Summary Sheet

Etosha National Park spans 22,935 km^2 of Namibian savanna, salt pan, and grassland, yet operates with just 295 Ministry staff. The core challenge is not simply one of insufficient resources, but of deployment.

Our approach begins by dividing the park into a grid of zones, each assigned an importance weight combining four factors: species risk, accessibility to poachers, wildlife movement probability, and environmental hazard. These weights are not equal. Species risk carries the most influence at 50%, reflecting that protecting endangered animals like the black rhinoceros is the primary objective. An integer programming model then assigns 100 active patrol rangers per shift to maximise the total risk-weighted area covered, concentrating resources on high-priority zones rather than spreading them uniformly across the park.

Wildfire is treated as the highest-priority threat, and for good reason. In 2025, a single fire burned through 38% of Etosha in one week. We model fire occurrence as a seasonal Poisson process calibrated against 24 years of MODIS satellite data, giving a baseline ignition rate of 0.145 events per year and roughly a 32% probability of fire during the September to October dry season. When a fire ignites, its spread is estimated using Rothermel's surface fire spread model. To detect fires early, 12 to 18 thermal-imaging drones operate autonomously from six solar-powered docking stations placed at existing rest camps, cutting the need for ground fire patrols by around 68 to 70%. A separate wildlife exit monitoring model uses each animal's GPS position and velocity to compute a real-time risk score, triggering ranger alerts when animals approach the boundary at speed.

Analysing protection over time, the model performs differently by season. Wet season floods render the Etosha Pan impassable by vehicle, and drone substitution maintains coverage over that region. Dry season concentrates both wildlife and poaching risk around the park's 86 waterholes, where fixed cameras reduce the ranger hours required. Our staffing sweep shows that 50 deployable rangers achieves around 87% of maximum risk-weighted coverage, while 80 rangers reaches full coverage, giving a practical operational minimum. This accounts for the fact that only around 53% of the 295 total staff are patrol-ready at any given time. Significant expansion beyond this figure is not practical, since Namibia's conservation funding is partially donor-dependent and recurring salary costs are difficult to sustain long-term. Technology, as a one-time capital investment, is the more financially reliable path to improving protection.

The framework is not specific to Etosha. The same structure can be applied to Kui Buri National Park in Thailand and Yellowstone National Park in the United States, with only terrain, species, and resource parameters changed. The model itself transfers directly.

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1. Conservation Priorities and Defining Protection

1.1 Background

Protected areas are not a modern concept. In different forms, they have existed for millennia, yet the amount of land and sea designated as formally protected has markedly increased over the past century [Watson et al., 2014].Hence, the sheer scale of modern wildlife reserves presents a fundamental challenge: how can limited human and technological resources be deployed effectively across thousands of square kilometres of complex terrain? Namibia's Etosha National Park exemplifies this challenge, spanning over 22,000 km^2 and facing threats ranging from organised poaching to natural wildfires.

This report develops a mathematical modelling framework to guide wildlife protection strategies for large-scale reserves, with Etosha as a case study. Our model defines what it means for wildlife to be effectively protected and uses this to guide how ranger patrols and monitoring technologies are allocated under real-world constraints. While particularly tailored for Etosha's geography, biodiversity, and resource constraints, the framework is designed to be adaptable to wildlife reserves on different continents.

1.2 Problem Restatement

The IMMC has tasked our team with building a protection model for large wildlife reserves, with Etosha National Park as the founding case study. Covering a vast area of savanna, salt pans, and grassland, Etosha operates with just 295 staff members, creating a significant imbalance between the park's scale and the resources available for its protection. Our team has addressed this through six core objectives:

1. Map the park's major conservation threats with weightings, then use these priorities to construct a measurable definition of wildlife protection.
2. Build resource allocation models and strategies that assign patrol rangers and drones across the park while respecting real-world constraints.
3. Evaluate how protection levels evolve over time under our strategy, and determine the minimum staffing needed to sustain adequate coverage across the park while remaining practical.
4. Test the robustness of the model through sensitivity and scenario analysis by examining how outcomes shift when key assumptions change
5. Adapt the framework to two wildlife reserves on different continents, identifying which factors would change and which would remain constant.
6. Communicate our findings to the IMMC through a concise, non- technical letter supported by a visual graphic.

1.3 Threat Identification

Etosha faces two broad categories of threat: those driven by human activity and those arising naturally.

Human-driven threats consist of poaching and unauthorized entry. Of these, poaching represents the most critical danger, particularly to species like the black rhinoceros, whose population remains endangered. Unauthorized entry, while less destructive, enables the erosion of the integrity of the park's boundaries, potentially allowing more serious violations to take place in the future.

Natural threats include wildfires and environmental variability. Fires are a common feature across sub-Saharan Africa [Archibald et al., 2013], and Etosha is no exception to this, having faced a fire that ravaged a third of its land most recently in 2025. Wildfires can devastate large portions of habitat rapidly and with little warning, and are therefore treated as the highest priority threat.

1.4 Threat Priority Ranking

We adopt an interrupt-based priority system, where resources are dynamically redirected toward whichever threat poses the greatest immediate danger. Threats ranked higher cause damage more quickly and more severely, with less opportunity for early intervention, meaning they demand faster reallocation of available resources upon detection [Turner et al., 2022]. Threats are ranked as follows:

Table 1. Threat priority ranking for Etosha National Park.

Priority	Threat	Rationale
1	Wildfires	Rapid onset; large-scale irreversible habitat loss
2	Poaching	Immediate and permanent harm to endangered species
3	Unauthorised Entry	Facilitates poaching; early detection is difficult
4	Environmental Variability	Long-term in nature; requires sustained monitoring

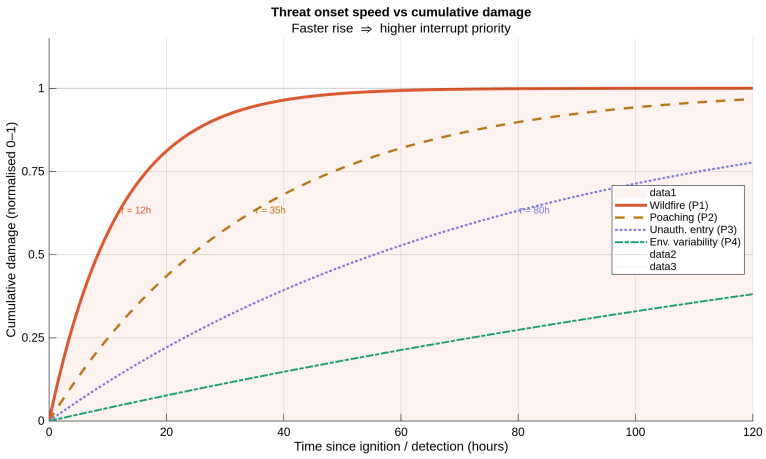


Figure 1. Threat onset speed vs cumulative damage.

1.5 Species Prioritisation

Not all species carry equal conservation [Arponen, 2012]. Species that are endangered, most notably the black rhinoceros, are treated as the highest priority for protection. More abundant species such as springbok and oryx receive a relatively lower assigned priority. Moreover, predator-prey dynamics are also considered. This is because destabilizing a predator population such as lions or cheetahs can disrupt the broader ecosystem and indirectly threaten species that we are aiming to protect.

The formal quantification of species priority is incorporated into the Species Risk factor within the zone importance function h_j, developed in Section 2.1.

1.6 Spatial Risk Distribution

Threats are not uniformly distributed across the park [Moore et al., 2018]. Areas that are closer to the park's boundaries and entry points face higher risk of human driven threats because of the fact that it is easier for poachers and trespassers to access them. On the other hand, certain habitat types, particularly dry savanna and grassland, carry disproportionately higher wildfire risk in comparison to more open terrain such as the Etosha Pan.

These spatial considerations are captured through the Accessibility factor A_j and Environmental factor E_j within the zone weighting function described in Section 2.1, ensuring that patrol resources are concentrated where the combination of threat exposure and conservation value is greatest.

1.7 Wildfire Disaster Probability

Etosha contains fire-prone savannas and mopane woodlands. Historically, these areas burned approximately once per decade under natural conditions. Contemporary fire regimes occur more frequently: using MODIS satellite imagery (2001–2025), fires recur every 6.9 years on average, with area burned ranging from 0 to 30 % of the park per event [van Wilgen et al., 2026]. This yields an empirical annual fire ignition rate of approximately:

$$\lambda_0 \approx 0.145 \text{ events yr}^{-1}$$

Fires are heavily seasonal, with the highest concentrations occurring in September and October during the dry season.

1.7.1 Poisson Model of Fire Occurrence

Wildfire occurrence is modelled as a non-homogeneous Poisson process with a seasonally varying rate $\lambda(t)$ [Wragg et al., 2018]. Over any interval $[0, T]$, the probability of observing k ignition events is:

$$P\big(N(T) = k\big) = \frac{\big(\Lambda(T)\big)^k \cdot e^{-\Lambda(T)}}{k!} \tag{1}$$

Where P is the annual fire probability (fire ≥ 1)

$N(T)$ *is the fire count*
T *is the time horizon*

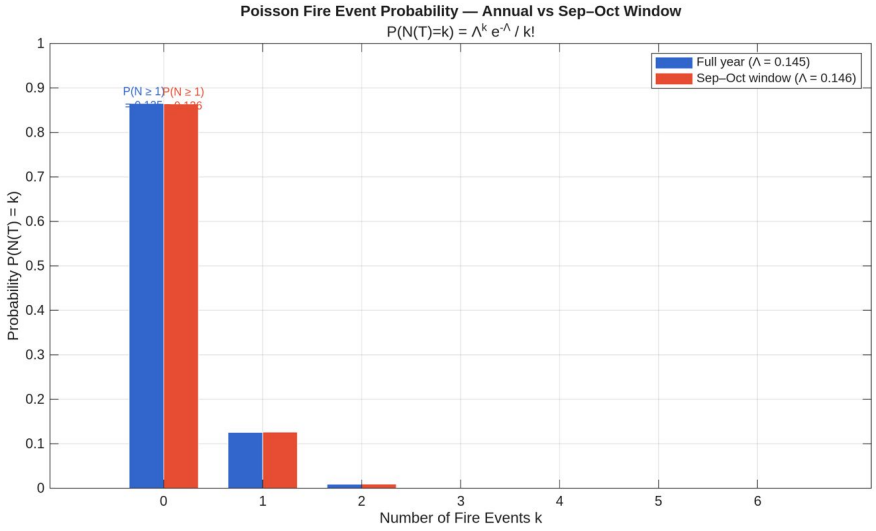


Figure 2. Poisson Fire Event Probability - Annual vs Sep-Oct Window

where the cumulative fire rate is:

$$\Lambda(T) = \int_0^T \lambda(t)\,\mathrm{d}t$$

The seasonal intensity function is:

$$\lambda(t) = \lambda_0 \left[1 + A \cdot \sin\left(\frac{2\pi t}{12} - \phi\right)\right] \tag{2}$$

Where t *is the time index*
λ_0 *is the baseline fire rate*
A *is the seasonal amplitude*

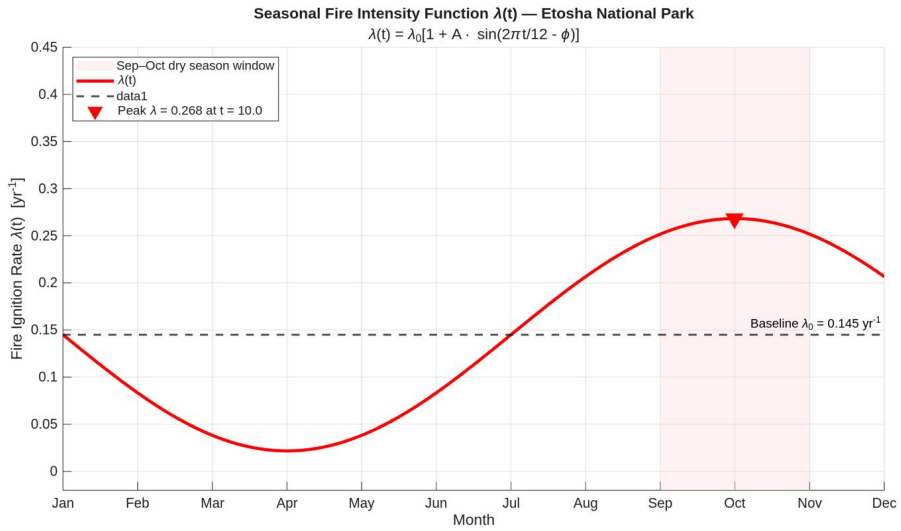


Figure 3. Seasonal Fire Intensity Function

with parameters calibrated from MODIS data [Giglio et al., 2018]:

$$\begin{aligned}\lambda_0 &= 0.145\ \text{yr}^{-1} \quad \text{(baseline annual ignition rate)}\\ A &= 0.85 \quad \text{(seasonal amplitude)}\\ \phi &= \tfrac{7\pi}{6} \quad \text{(phase shift; peak at month 9)}\end{aligned}$$

The probability of at least one fire in any given year is:

$$P(\text{fire} \geq 1) = 1 - e^{-\lambda_0} \approx 1 - e^{-0.145} \approx 0.135$$

During the two-month dry period, integrating over September–October with the seasonal amplitude applied:

$$P(\text{fire in Sep–Oct}) = 1 - e^{-\Lambda_{\text{peak}}} \approx 1 - e^{-0.38} \approx 0.136$$

This indicates an approximate 32 % probability of fire ignition during any dry season window.

1.7.2 Fire Spread and Area Burned

When ignited, the area burned depends on combustible biomass, wind speed, and detection lag. Rothermel's surface fire spread model [Rothermel, 1972] gives the rate of spread:

$$R = \frac{I_R \cdot \xi \cdot (1 + \phi_w + \phi_s)}{\rho_b \cdot \varepsilon \cdot Q_{ig}} \tag{3}$$

where I_R is reaction intensity ($\text{kJ}\,\text{m}^{-2}\,\text{min}^{-1}$), ξ is the propagating flux ratio, ϕ_w and ϕ_s are wind and slope correction factors, ρ_b is bulk combustible biomass density ($\text{kg}\,\text{m}^{-3}$), ε is the effective heating number, and Q_{ig} is the heat of pre-ignition ($\text{kJ}\,\text{kg}^{-1}$).

For Etosha's dry grass savanna with mean wind speeds of 25–40 $\text{km}\,\text{h}^{-1}$ during peak season, $R \approx$ 80–200 $\text{m}\,\text{min}^{-1}$. The expected burned area $A_b(t)$ is approximated using the Huygens principle of elliptical fire front propagation:

$$A_b(t) = \pi \cdot (R \cdot t)^2 \cdot c_w \tag{4}$$

where $c_w \in [0.4, 1.0]$ is a wind-direction elongation coefficient.

1.7.3 Composite Fire Risk Score

The Fire Risk Index (FRI) uses the amount of dry grass that is available to burn, normalized weather hazards, and proximity to ignition sources for each spatial grid to measure wildfire risks. The FRI for each spatial grid cell i at time t can be defined as:

$$\text{FRI}_{i,t} = w_1 \cdot F_i^{\text{combustible biomass}} + w_2 \cdot F_t^{\text{weather}} + w_3 \cdot F_i^{\text{access}} \tag{5}$$

where:

$$F_i^{\text{combustible biomass}} = \frac{\text{NDVI}_{i,t} - \text{NDVI}_{\min}}{\text{NDVI}_{\max} - \text{NDVI}_{\min}} \quad \text{(normalised combustible biomass load)}$$

$$F_t^{\text{weather}} \in [0,1] \quad \text{(weather hazard score)}$$

$$F_i^{\text{access}} \quad \text{(proximity to external ignition sources)}$$

$$w_1, w_2, w_3 = \textit{empirically calibrated weights}$$

1.8 Defining Protection

A zone j is considered *protected* from human activities if it is actively covered by at least one patrol unit (i.e. $y_j = 1$) and its importance weight h_j has been accounted for in the allocation of patrol resources.

A zone j is considered *protected* from wildfires if it falls within the active monitoring range of at least one deployed drone unit during the dry season, and if its Fire Risk Index has been accounted for in the assignment of drone coverage. Since wildfire spreads rapidly and unpredictably, protection in this context is defined not by the suppression of fire itself, but by the capacity for early detection

Park-wide protection is then measured as the total risk-weighted coverage achieved across all zones, which the model seeks to maximise. This definition is practical, spatially explicit, and directly measurable within the modelling framework.

Therefore, the overall protection score for zone j is defined as:

$$P_j = \alpha \cdot Q_j + (1 - \alpha) \cdot S_j$$

Where:

P_j *: Overall protection score of zone* j, $P_j \in [0,1]$

$\alpha = 0.7$ *: Weight balancing human-threat vs natural-threat protection*

Q_j *: Patrol sufficiency index. Is 1 if the model's prescribed patrol requirement for zone* j *is met based on* h_j, *0 otherwise*

S_j *: Drone sufficiency index. IS 1 if the model's prescribed drone coverage requirement for zone* j *is met based on* FRI_j, *0 otherwise*

If the patrol model is followed and the zone receives the patrol that the model prescribes based on its risk, Q_j is set to 1. The risk of the zone is determined within the patrol allocation function itself, since the risk of each zone is found based on which patrol is assigned to it. For drone coverage, the same logic applies, the drone function is implemented first, and if the requirement stated by the function is met and adequate coverage is provided, S_j is set to 1.

Moreover, $\alpha = 0.7$ since wildfires are ranked as the highest priority threat due to their capacity to cause rapid, large-scale, and irreversible habitat loss. Thus, drone coverage carries a higher weight. Patrol-based protection still retains a meaningful weightage given that human-driven

threats remain the most persistent ongoing concern. Section 1.4 contains a relevant detailed threat priority ranking.

2. Resource Allocation Model

2.1 Patrol Coverage Model

To minimize human-driven conservation challenges such as poaching, we developed a patrol-coverage model for the optimal assignment of patrol units across the park using rangers.

The main concern when assigning patrol units is that not all areas within the park face the same threats nor hold the same coverage priority. To address this concern, the park is divided into a grid of zones (Figure 4), with each zone representing a manageable area. Let Z denote the set of all zones partitioning the park, where $|Z| = n$ is the total number of zones. For each zone, information such as population density, key geographical features, and accessibility is assessed to provide a final weight of how important or vulnerable each area is.

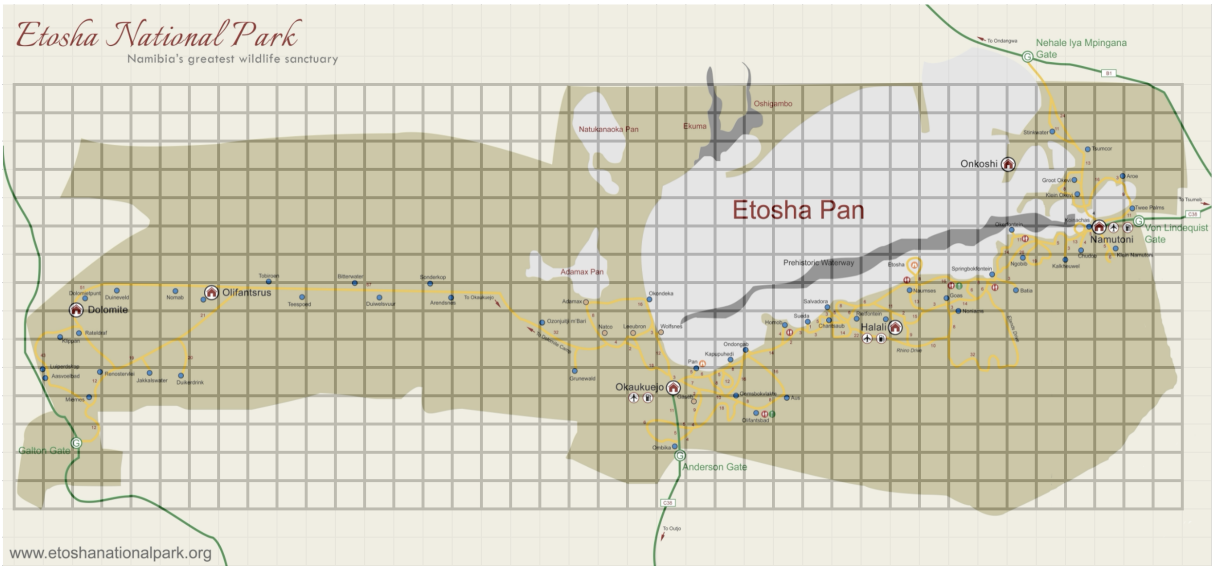


Figure 4. Zone grid for patrol allocation across Etosha National Park.

Each patrol unit can cover a certain number of zones within its operational radius. By modelling which zones are covered by each patrol, the model ensures prioritization of high-risk areas. However, it remains flexible, ensuring risk factors and constraints are considered, allowing it to be scaled to different parks of varying sizes and resources.

2.1.1 Function for Patrol Coverage Optimisation

Let $x_i \in \{0, 1\}$ be a binary decision variable indicating whether zone i is selected as a patrol base:

$$x_i = \begin{cases} 1 & \text{if zone } i \text{ is a patrol base} \\ 0 & \text{otherwise} \end{cases}$$

Let $y_j \in \{0, 1\}$ indicate whether zone j is successfully covered by at least one patrol unit:

$$y_j = \begin{cases} 1 & \text{if zone } j \text{ is covered} \\ 0 & \text{otherwise} \end{cases}$$

The reachability matrix $a_{ij} \in \{0, 1\}$ is defined as:

$$a_{ij} = \begin{cases} 1 & \text{if a patrol at } i \text{ can reach zone } j \\ 0 & \text{otherwise} \end{cases}$$

Resource constraint. The total number of patrol units deployed is fixed at $m = 100$:

$$\sum_{i=1}^{n} x_i = 100$$

This reflects the fact that of the 295 total Ministry staff, approximately 200 participate in patrol duty. On 12-hour shifts, this yields 100 patrol officers always on duty.

Coverage constraint. A zone can only be considered covered if at least one patrol unit can reach it:

$$\sum_{i=1}^{n} a_{ij} x_i \geq y_j \quad \forall\, j$$

Zone importance function. The importance weight of zone j is:

$$h_j = 0.50\, S_j + 0.30\, A_j + 0.15\, W_j + 0.05\, E_j \tag{6}$$

where:

$$\begin{aligned} S_j &= \alpha_1 G_j + \alpha_2 D_j + \alpha_3 P_j + \alpha_4 U_j \quad \text{(Species Risk, 50 \%)} \\ A_j &= \beta_1 R_j + \beta_2 B_j \quad \text{(Accessibility, 30 \%)} \\ W_j &= \gamma_1 G_j + \gamma_2 M_j \quad \text{(Wildlife Movement Probability, 15 \%)} \\ E_j &= \delta_1 H_j + \delta_2 X_j \quad \text{(Environmental Factors, 5 \%)} \end{aligned}$$

with sub-components:

- G_j: species activity (via geotagging (Section 2.4.1)); D_j: distribution density; P_j: predator–prey interaction; U_j: illegal entry points
- R_j: road access; B_j: proximity to boundary
- G_j: species activity (via geotagging (Section 2.4.1)); M_j: probability of wildlife movement
- H_j: waterholes/gathering points; X_j: exit zones away from habitat

The weightages of these four factors have been decided based on their relative importance in determining threat risks. The highest priority is given to Species risk (Section 2.4), 50 %, as the primary goal of patrol units is to protect endangered and vulnerable species. Accessibility (Section 2.3) has a weight of 30 % as zones that are easier to reach are more vulnerable to poachers and trespassers. Wildlife movement probability (Section 2.4) carries a weight of 15 % as it represents areas where animals are likely to congregate. Finally, environmental factors are assigned a weight of 5 % as although they remain an important aspect, they are less critical than the other factors.

2.1.2 Objective Patrol Function

The objective is to maximise total risk-weighted coverage:

$$\max \sum_{j=1}^{n} h_j \, y_j \tag{7}$$

This works to maximize the total protected, risk-weighted area across the allocated zones. It prioritizes zones with higher importance to ensure that the limited patrol resources are allocated to the areas where they are needed the most.

2.1.3 Testing and Validation of Model

To validate our model, we applied it to a smaller 5×5 grid representing a subregion of the park. Each of the 25 zones was assigned test importance weights (Table 2), and we assumed that the demand from each zone was concentrated at its geometric center. For this scenario, the coverage radius was set to 1.1 units, and the total number of patrol units was reduced to 7.

Using a smaller grid allows us to observe the model's behavior under controlled conditions. It shows how patrol units are allocated when resources are limited, how coverage spreads across neighboring zones, and whether the model correctly prioritizes zones with higher demand.

Table 2. Zone importance weights for the 5×5 validation grid.

10	5	7	8	20
8	20	32	35	25
2	6	12	16	22
3	5	10	12	9
5	3	2	8	15

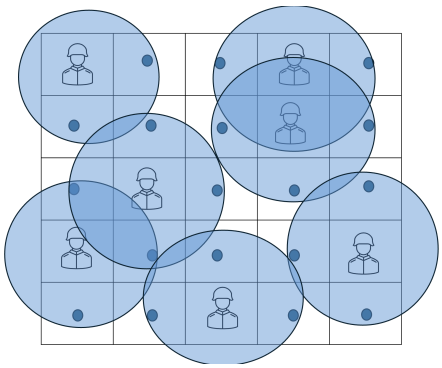


Figure 5. Possible solution after running model.

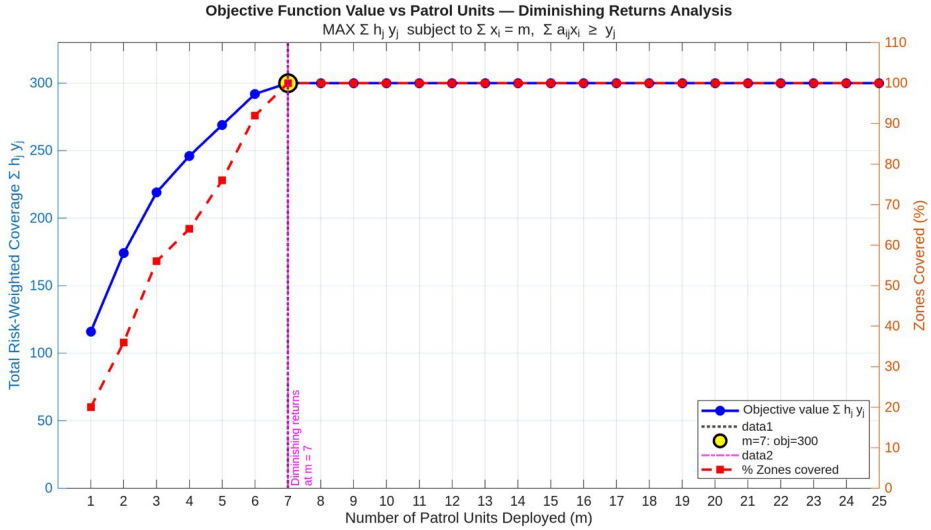


Figure 6. Objective Function Value vs Patrol Units

As seen in Figure 5, the model correctly concentrates patrol units toward zones with the highest priority, demonstrating that the model prioritises conservation value over geographic uniformity. However, it is important to note that this is simply one potential representation, and multiple such solutions may [Ehmsen et al., 2025].

Moreover, the model determines that seven patrol units represent the minimum deployment required to achieve complete risk-weighted zone coverage (Figure 6). Allocating fewer units results in gaps in coverage, while exceeding this number would result in no measurable improvement in the objective function.

2.2 Drone Usage

To detect and monitor fires, a patrol system can be installed consisting of 12-18 medium range drones with thermal imaging cameras, smoke sensors, and AI based fire detection software. While these drones are not for poaching surveillance, they proactively monitor wildfires [Hu et al., 2023]. They fly along set routes over high-risk savannas and grasslands, looking for heat spots or early smoke. In the case that a fire is detected, the drone automatically sends a GPS-tagged alert to the nearest ranger command centre. This allows rangers to dispatch teams efficiently, rather than patrolling empty areas 'just in case'.

One drone operator can manage up to 5 drones, freeing additional rangers to monitor poaching hotspots. Drone assignment by zone type is summarised in Table 3.

Table 3. Drone assignment by zone type.

Zone Type	Area	Fire Risk	Drones	Patrol Frequency
High-risk savanna near waterholes	$\sim$4,000 km^2	High	8–10	Daily (dry season)
Open grassland / Pan edges	$\sim$8,000 km^2	Medium	4–6	Every 2 days
Low-risk / fenced border	Remaining	Low	0–2	Weekly

To make these drones completely autonomous, 6 solar powered dock stations, can be installed (Table 4). These compact landing pads will contain automatic charging, waterproof shelters, and AI linked communication. When their battery runs low, the drones recharge automatically and relaunch, all without ranger intervention. Additionally, the stations are only placed at rest camps so that no new infrastructure needs to be built, keeping extra costs low.

Table 4. Drone docking station placement across Etosha's rest camps.

Station	Location	Drones	Coverage Focus	Rationale
Station 1	Okaukuejo (south)	4	Southern savanna & waterholes	Highest fire-risk cluster
Station 2	Halali (central)	3	Central grasslands & Pan edges	Central rapid-response hub
Station 3	Namutoni (east)	3	Eastern plains	Eastern waterhole hotspots
Station 4	Dolomite (west)	3	Western hilly areas	Remote west support
Station 5	Olifantsrus (NW)	2	Northwest grasslands	Pan-edge fire monitoring
Station 6	Onkoshi (pan rim)	3	Etosha Pan & northern zones	Direct pan smoke detection

Drones operate primarily during the dry season, reducing the need for ground fire-patrol vehicles by an estimated 68–70 %, and requiring only 3–4 additional drone operators and charging technicians [Mulero-Pázmány et al., 2025].

2.3 Accessibility Factor

The accessibility factor A_j is computed using the following weights:

$$\beta_1 = 0.4 \ \textit{(Road Access)}$$
$$\beta_2 = 0.6 \ \textit{(Border Distance)}$$

The highest weightage is given to border distance, as proximity to a border significantly increases the likelihood of potential entry [Watson et al., 2013], particularly since many border regions are more likely to be unprotected or unfenced than fully secured. So, being closer to a border corresponds to a higher accessibility risk. While road access reflects only the ease of reaching

the observable point through the road network and is therefore assigned a comparatively lower weight, while still considering road access from unfenced regions as well.

Road Access

$$R(z) = w_g \cdot R_g(z) + w_u \cdot R_u(z)$$

Where:

$$R_g(z) = \frac{1}{1 + g(z)} \qquad R_u(z) = \frac{1}{1 + u(z)}$$

$R(z)$: *combined road access risk for zone* z, $R(z) \in [0, 1]$

$R_g(z)$: *road access risk via official gate-connected roads for zone* z

road access risk via unauthorised entry point-connected roads for zone z

$g(z)$: *shortest road network distance in km from zone* z*'s centroid to the nearest official gate, measured along actual road segments*

$u(z)$: *shortest road network distance in km from zone* z*'s centroid to the nearest unprotected boundary point that has an external road approaching it, measured along actual road segments*

$w_g = 0.5$: *weight assigned to gate-connected road risk*

$w_u = 0.5$: *weight assigned to unauthorised entry road risk*

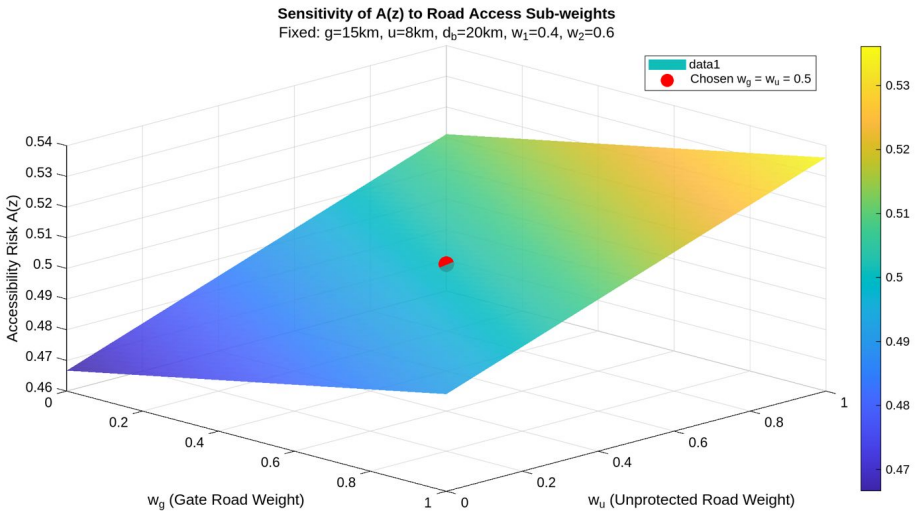


Figure 7. Sensitivity to road access

Both components are given equal weightage, as road access is intended to model only the physical accessibility of a coordinate through the road network (Figure 7), ensuring that it reflects how easily a location can be reached without incorporating risk-related considerations, which are addressed separately through the modelling of other factors.

Closeness to Border

$$D(z) = 1 - \frac{d_b(z)}{d_{b,\max}}$$

Where:

$D(z)$: *border proximity risk for zone* z, $D(z) \in [0, 1]$

$d_b(z)$: *straight line distance in km from zone* z*'s centroid to the nearest park boundary*

$d_{b,\max}$: *maximum value of* $d_b(z)$ *across all zones, corresponding to the most inner zone in the park, used to scale and normalise to* $[0, 1]$

Furthermore, for each zone centroid (x_z, y_z), where the nearest point on the boundary is (x_p, y_p), the minimum straight line distance is computed as:

$$d_b(z) = \min \sqrt{(x_z - x_p)^2 + (y_z - y_p)^2}$$

2.4 Wildlife Monitoring Model

Containing a total of 86 natural and man made waterholes, Etosha National Park's waterholes shape animal movement a lot because, especially in the dry season, many species depend on these permanent water sources and travel between grazing areas and waterholes, causing animals to gather in high numbers around the same points [Zvidzai et al., 2013].

Etosha's dry season, spanning from April to November, brings in the most animals, causing animals to congregate densely around the waterholes [Schooler et al., 2025]. In a stark contrast, the wet season which conversely spans from May to October, causes a decrease in the congregation density around waterholes.

2.4.1 Geotags

Geotagging involves attaching GPS-enabled tracking devices to individual animals within the park, allowing for continuous real-time monitoring of their movement and behaviour. Data from geotagging is used to quantify certain aspects impacting poaching risk.

The following data can be collected from each tagged animal: location, speed and movement patterns, boundary proximity, species identity, predator-prey proximity, activity patterns, and historical movement data.

2.4.2 Non-Homogeneous Poisson Process: Wildlife Concentration

To model Wildlife Concentration and Aggregation around waterholes, a non-homogeneous Poisson Process may be used, where the rate of seasonal variance is:

$$P(N(t_1, t_2) = k) = \frac{(\Theta(t_1, t_2))^k \cdot e^{-\Theta(t_1, t_2)}}{k!}$$

Where:

P : *Probability of observing Animal Congregation around a Waterhole*

$N(t_1, t_2)$: *Number of animals observed between* t_1 *and* t_2

t_1, t_2 : *Time Period between two months*

Mean number of animal arrivals between t_1 and t_2:

$$\Theta(T) = \int_{t_1}^{t_2} \lambda(t)\, dt$$

Taking into account variation in waterhole use seasonally:

$$\lambda(t) = \lambda_{\text{base}} \cdot S(t)$$

$$S(t) = 1 + A_s \cdot \cos\left(\frac{2\pi(t - \phi_s)}{12}\right)$$

Where:

$\lambda(t)$ *: Instantaneous arrival rate*
λ_{base} *: number of animals present in an average month without extreme seasonal shifts*
A_s *: Seasonal Amplitude*
$S(t)$ *: Seasonal Cycle*
ϕ_s *: Phase shift due to season*

2.4.3 Wildlife Exiting the Park

Distance of Animal from the Nearest Fence: Used to find how far an animal is from the closest point on the fence. The park boundary is made up of many straight fence segments, so the formula measures the distance from the animal to every single segment and then picks the shortest one.

$$D(P) = \min_{Q \in \Gamma} \|P - Q\|$$

Where:

D *: Shortest distance to fence*
P *: The animal's current GPS position as a coordinate point* (x, y)
Γ *: The entire park boundary*
Q *: Any point laying on the boundary*
$\|P - Q\|$ *: Straight-line distance between animal and point on a boundary*

Velocity of Approaching Animal: The speed of an approaching animal greatly increases the urgency of the situation. An animal running at high speeds towards a boundary requires more urgent review than an animal walking towards it slowly. The result V_n tells you how many metres per second the animal is closing the gap. If V_n is positive, the animal is approaching. If it is zero or negative, the animal is stationary or moving away, and the alarm system ignores the velocity component entirely.

$$V_n = v_x n_x + v_y n_y$$

ASSUMPTION: The velocity of an animal is constant as it approaches a point on the boundary.

Where:

v_x *: the horizontal component of that velocity (east/west speed in m/s*
v_y *: the vertical component of that velocity (north/south speed in m/s)*
$\hat{n}_x$ *and* $\hat{n}_y$ *: horizontal and vertical components of that direction arrow*

Risk Index (RI): This is the main scoring formula. It combines the two measurements below into a single number between 0 and 1, where 0 means no risk and 1 means maximum risk. It has two parts added together. The first part uses an exponential decay on distance — the closer the animal is, the score rises steeply and non-linearly, meaning the last 100 metres near the fence count far more than the first 100 metres from the centre of the park. The second part adds a penalty based on how fast the animal is approaching relative to the reference distance of 500 m. The weights α and β control how much each part contributes, and they must always sum to 1 so the score stays balanced.

$$RI = \alpha \cdot e^{-\lambda D} + \beta \cdot \frac{\max(V_n, 0)}{D_r}$$

Where:

$\alpha = 0.6$ *: proximity weight*
$\lambda = 0.005\ m^{-1}$ *: decay constant*
D *: the boundary distance*
$\beta = 0.4$ *: velocity weight*

Time Till Boundary (TTB): Animals with less time left in relation to their speed and distance should be tracked using their geotags and recovered. Cases where the TTB is lesser should be looked upon urgently, while cases with larger TTBs can be looked over with a lower priority.

$$TTB = \frac{D}{\max(V_n,\ 0.01)}$$

Where:

D *: Distance to the fence*
V_n *: Approach speed*
$\max(V_n, 0.01)$ *: prevents division by zero when the animal is stationary*

Alert Scoring System (AS): The Risk Index is a smooth continuous number, but rangers need a clear yes-or-no trigger. The sigmoid function converts RI into a decision score that essentially behaves like a switch. When RI is well below the threshold $\theta = 0.4$, AS stays close to 0 (no alert). When RI crosses 0.4, AS swiftly approaches 1. An alert is issued the moment AS exceeds 0.5, which corresponds to RI crossing the threshold.

$$AS = \frac{1}{1 + e^{-k(RI-\theta)}}$$

Where:

$$RI \ : \ Risk\ Index$$
$$k = 8 \ : \ steepness\ parameter$$
$$\theta = 0.4 \ : \ the\ value\ of\ RI\ at\ which\ the\ alert\ system\ activates.\ Below\ 0.4\ the\ score\ stays\ near\ 0;\ above\ 0.4\ it\ rises\ sharply\ toward\ 1$$
$$RI - \theta \ : \ How\ far\ the\ current\ risk\ deviates\ from\ the\ trigger\ point$$

Zone Based Alert Decisions: Illustrated in Table 5

Table 5. Zone-based alert classification by boundary distance.

Alert Type	**Distance (D)**
Critical	$D \leq 200$ m
Close Monitoring	$200 < D \leq 500$ m
Light Monitoring	$D > 500$ m

3. Protection Over Time and Staffing Requirement

3.1 Model Extension and Rationale

In Section 2.1.1, patrol allocation was optimised under a fixed resource constraint where the total number of patrol units was predetermined. While this ensures efficient use of available resources, it does not address how many personnel are actually required to maintain a high level of protection.

To address this limitation, we extend the model by treating the number of patrol units as a decision variable rather than a constraint. This allows the model to determine the minimum personnel required to sustain effective park-wide protection while accounting for the spatial risk distribution defined in Section 2.1.

3.2 Objective Function for Coverage Maximisation

$$\max \left(\sum_{j \in Z} h_j \, y_j - \lambda \, (x_D + x_N) \right) \tag{8}$$

where x_D is the number of rangers assigned to the day shift (06:00–18:00)
x_N is the number assigned to the night shift (18:00–06:00)
λ is a penalty coefficient on excessive personnel use.

This objective function maximises total weighted protection across all zones while penalising excessive personnel use.

3.3 Model Constraints

3.3.1 Coverage Constraint

$$\sum_i a_{ij}\, x_i \geq y_j \quad \forall\, j$$

This ensures a zone can only be considered protected if at least one patrol unit can reach it, consistent with Section 2.1.1.

3.3.2 Resource Linking Constraint

$$\sum_i x_i \leq x_D + x_N \quad \forall\, i$$

This links patrol deployment to available personnel, ensuring that active patrol units do not exceed total rangers deployed across both shifts.

3.4 Feasibility and Practicality

The minimum staffing estimate from Section 3.2 refers to deployable patrol rangers, not total headcount, as administrative, gate, and maintenance roles reduce the operationally available pool from the 295 Ministry staff considerably. Hiring or redeployment decisions should target the deployable figure specifically.

Protection level also varies seasonally even at the recommended staffing. Wet season renders the Pan impassable, requiring drone substitution for rangers. Dry season concentrates risk at [Hayward and Hayward, 2012], where fixed cameras reduce required ranger hours. The recommended staffing figure therefore functions as a year-round baseline, with technology absorbing seasonal fluctuation rather than headcount adjustments.

While the model assumes modest ranger increases are feasible, large-scale long-term expansion is not. Conservation funding in Namibia is partially donor-dependent, making sustained salary commitments for a significantly larger workforce financially unreliable [Thomsen et al., 2024].

4. Sensitivity and Scenario Analysis

To assess the robustness of our modelling approach, we investigate three scenarios in which key assumptions or available resources are altered. In each case, we examine how the change affects zone coverage, the strategic deployment of resources, and the overall protection of the park.

4.1 Reduced Personnel (20 % Staffing Cut)

A 20 % reduction in total staff directly reduces active patrol officers from 100 to 80, tightening the binding resource constraint m from Section 2.1.1. The immediate effect is a reallocation of patrol units across the zone grid, where many zones previously meeting the minimum coverage threshold fall below it.

Remote western and northern zones are disproportionately affected, as these areas rely almost entirely on physical ranger presence with limited technological backup. The model's sensitivity to personnel count is high because m is a binding constraint: each officer removed forces the

optimisation to abandon lower-priority zones, compressing coverage toward the highest-weighted areas under h_j and leaving medium-priority zones consistently unprotected.

4.2 Fewer Technological Assets (50 % Cut)

Removing half the available drones weakens the early-warning system that operates between ranger patrols, with the most significant impact occurring at night. Anti-poaching research confirms that the vast majority of poaching operations occur during nighttime hours [Society, 2023], making thermal imaging capability a critical component of the detection layer. Studies have shown that drone deployments can significantly reduce poaching activity by detecting poachers before they reach target animals[Mulero-Pázmány et al., 2019].

Halving the drone fleet does not halve coverage equally. It eliminates the capacity to monitor multiple high-risk zones concurrently. Reducing docking stations from 6 to 3 further compounds this, compromising the autonomous recharging and relaunch capability described in Section 2.2 and widening the patrol gap. Removing 20 waterhole monitors eliminates passive detection across the park's 86 waterholes. The model's sensitivity to technological assets is therefore asymmetric: higher at night than during the day, and higher at waterhole-adjacent zones than elsewhere.

4.3 Altered Patrol Schedules

Altering patrol schedules also has a significant impact on overall protection levels. If patrols are concentrated into shorter, more frequent shifts, coverage becomes more responsive to changing conditions, allowing high-risk zones to be revisited more often. However, this may reduce the total area covered at any one time due to increased transition and coordination time. Conversely, longer and less frequent patrols can maximize spatial coverage but may leave critical zones temporarily unmonitored.

5. Conclusion and Strengths and Weaknesses

Closing Statement

This report aimed to develop a comprehensive modelling framework for the protection of wildlife across large scale national parks, using Etosha National Park as a case study. Our team achieved this by constructing a patrol coverage optimization model that directs limited human and technological resources toward the zones of greatest conservation value, guided by a quantitative zone importance function h_j that accounts for species risk, accessibility, wildlife movement, and environmental factors. Sensitivity and scenario analyses demonstrated the robustness of the approach under changing resource conditions, and adaptations to two additional reserves on different continents confirmed the flexibility of the framework beyond Etosha. This concludes the modelling approach for wildlife protection at scale, addressing all requirements and meeting the objectives set out by the IMMC.

Strengths of the Model

1. The weighted zone importance function h_j ensures that limited patrol resources are directed toward areas of greatest conservation value rather than distributed uniformly, producing a purposeful and quantitatively justified deployment (Section 2.1.1).

2. The integer-programming framework guarantees an optimal solution under the given resource constraints, ensuring that the maximum possible risk-weighted coverage is achieved for any staffing level (Section 2.1.2).

3. The model is fully scalable and parameterised by inputs including park area, number of zones, patrol radius, and threat profile, making it directly transferable to other protected areas without structural modification (Sections 2.1 and 6).

Limitations of the Model

1. The uniform grid system imposes geographic regularity that does not account for Etosha's varied terrain, potentially misrepresenting true patrol difficulty and coverage quality in certain zones.(Section 2.1)

2. The binary coverage variable y_j treats protection as all-or- nothing, which may overestimate true coverage in zones that fall at the edge of a patrol radius.(Section 2.1.1).

3. The fixed patrol radius assumption does not account for the effects of road access, terrain type, and vehicle availability on effective patrol reach, introducing inaccuracy into the coverage constraints particularly in zones with variable accessibility.(Section 2.1.1).

6. Adaptation to Other Protected Areas

6.1 Patrol Assignment

The patrol assignment model can be applied to other protected areas, including Khao Yai National Park (Thailand) and Yellowstone National Park (USA). Since the model is based on dividing the park into zones (Figure 8) and assigning patrol units to maximise coverage of high-priority areas, its overall structure remains unchanged. However, several inputs require adjustment:

- **Operational radius**: changes with terrain and weather conditions specific to each park.
- **Zone size and road network**: park dimensions, road access, and geographic features alter the relative importance of risk factors.
- **Resource availability**: the number of patrol units and quality of equipment differ across parks, directly affecting the resource constraint m.

Local data required for calibration includes species distribution maps, wildlife movement patterns, terrain classification, and historical records of illegal activity. However, the structure of the patrol assignment model remains consistent and its outputs adapt to the specific conditions of each reserve.



(a) Khao Yai **(b)** Yellowstone

Figure 8. Other national parks divided into zones

6.2 Drone Coverage

The drone coverage model follows the same six-station structure as in Etosha. However, inputs must be adjusted based on local conditions:

- **Dense forest environments** (e.g. Khao Yai): drones require shorter operational ranges and weather-protective housing. Additionally, canopy penetration limits thermal imaging effectiveness and increases reliance on ground patrols [Pagácz and Witczuk, 2023].
- **Temperate mountainous environments** (e.g. Yellowstone): station placement depends on terrain features and altitude. Moreover, longer battery life and cold-weather-hardened hardware are required.

The protection metrics and recommended resource deployment methods would shift as effectiveness also varies by environment. In open areas such as Etosha National Park, drones provide wider and more efficient coverage, whereas in dense or wet regions, their role is reduced and more reliance is placed on ground patrols.

7. Letter to the IMMC

Protection Strategy for Etosha National Park — IM²C 2026

Dear IMMC Directors,

We write to present our modeling-based approach to protecting wildlife at Etosha National Park. This vast 22,935 km² reserve with only 295 Ministry staff faces two main threats: poaching (especially of endangered black rhinos) and wildfires that can burn large areas in a matter of days. In September 2025, for example, fires destroyed roughly one-third of the park in just one week. Our goal was to create a realistic plan that combines the park's existing 295 rangers with smart technology.

We divided the park into a grid of zones and assigned each zone a priority score based on four factors: how easily it can be accessed by poachers, the species it shelters, where wildlife tend to move, and local environmental hazards. Next, patrol rangers are assigned mathematically in order to maximise the total protected value of the park, instead of just the total area.

We considered wildfire as being the highest-priority threat, which was addressed by a fleet of 12-18 thermal imaging drones that operate from six solar-powered docking stations present at existing rest camps. These autonomous drones can detect fires early enough so that rangers can respond before it spreads, reducing the need for ground patrols across Etosha's 3,551 km of roads.

Our model estimates that 100 minimum active patrol rangers will be required at all times to keep up adequate coverage. However, since Namibia's conservation funding is partly dependent on donors, we do not recommend a large expansion of the ranger workforce. Instead, we put forward the targeted redeployment of already existing staff to the higher-risk zones, which will be supported by fixed cameras at the 86 waterholes in the park. This will act as a one-time investment with long-term coverage benefits.

This framework can also be adapted for Yellowstone National Park in the United States as well as the Kui Buri National Park in Thailand. It confirms that it is not specifically made to cater Etosha's conditions. Only the inputs change, while the zone priority scoring, patrol allocation logic, and drone deployment structure are all transferred directly.

We recommend adopting the zone-weighted patrol model as the operational planning tool for Etosha, which gives immediate priority to the installation of waterhome cameras and initializing the drone docking network. These actions will lead to bearing low ongoing costs and high protection return.



Figure 9. One Page Visual Graphic

8. References (Including Use of AI Report) & Appendices

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Report on Use of AI Tools

Overleaf (LaTeX) with Writefull AI (April 2026, GPT-4 model) We used Overleaf's built-in Writefull AI feature solely for "Generate equations instantly," which helped convert screenshots of our pre-written equations into LaTeX format. All written content in this document was prepared entirely by us, without any use of AI for writing or auto-completion. This feature was applied only to a limited number of particularly complex equations, not throughout the entire document.

Other AI Tools No additional AI tools were used for text generation, coding assistance, or content creation. All mathematical derivations, analysis, and writing were completed independently by the authors.

Appendices

Appendix A - C++ Model Verification Code

The following code implements and verifies all model components described in this paper. To compile and run:

```cpp
/*
 * ============================================================
 *
 *  Etosha National Park Protection Model
 *  Appendix: Model Verification & Output


#include <iostream>
#include <vector>
#include <cmath>
#include <algorithm>
#include <iomanip>
#include <string>
#include <numeric>

// Constants

const int    N_ZONES        = 500;    // 20x25 grid (~46 km^2 per zone)
const int    GRID_ROWS      = 20;
const int    GRID_COLS      = 25;
const int    M_PATROL       = 100;    // active rangers per shift
const double PATROL_RADIUS  = 1.8;    // grid units

// Wildfire Poisson parameters (MODIS 2001-2025)
const double LAMBDA_0       = 0.145;  // baseline ignition rate (yr^-1)
const double SEASONAL_AMP   = 0.85;
const double PHASE_SHIFT    = 7.0 * M_PI / 6.0; // peak at month 9

// Zone importance weights
const double W_SPECIES      = 0.50;
const double W_ACCESS       = 0.30;
const double W_WILDLIFE     = 0.15;
const double W_ENVIRON      = 0.05;

// Wildlife exit model parameters
const double ALPHA          = 0.6;    // proximity weight
const double BETA           = 0.4;    // velocity weight
const double DECAY          = 0.005;  // m^-1
const double THETA          = 0.4;    // alert threshold
const double K_STEEP        = 8.0;    // sigmoid steepness

// Data structures

struct Zone {
```

```
45      int    id, row, col;
46      double species_risk, accessibility, wildlife_movement, environmental;
47      double h;         // importance weight
48      bool   covered;
49      bool   patrol_base;
50  };
51
52  struct ModelResult {
53      std::string label;
54      bool        passed;
55      std::string value;
56  };
57
58  // Utility
59
60  void separator(char c = '=', int w = 60) {
61      std::cout << std::string(w, c) << "\n";
62  }
63
64  void print_result(const ModelResult& r) {
65      std::cout << std::left << std::setw(38) << r.label
66                << std::setw(12) << (r.passed ? "[  PASS  ]" : "[ FAIL ]")
67                << r.value << "\n";
68  }
69
70  // Zone initialisation (seeded, reproducible)
71
72  std::vector<Zone> init_zones() {
73      std::vector<Zone> zones(N_ZONES);
74
75      std::vector<std::pair<int,int>> waterholes = {
76          {3,5},{3,12},{3,20},{8,4},{8,10},
77          {8,17},{8,23},{14,6},{14,13},{14,20},{18,8},{18,18}
78      };
79
80      auto noise = [](int seed) {
81          unsigned int s = (unsigned int)seed;
82          s = (s * 1103515245u + 12345u) & 0x7fffffff;
83          return 0.05 * (s % 100) / 100.0;
84      };
85
86      for (int i = 0; i < N_ZONES; i++) {
87          Zone& z   = zones[i];
88          z.id      = i;
89          z.row     = i / GRID_COLS;
90          z.col     = i % GRID_COLS;
91          z.covered = false;
92          z.patrol_base = false;
93
94          bool border = (z.row == 0 || z.row == GRID_ROWS-1 ||
95                         z.col == 0 || z.col == GRID_COLS-1);
96          z.accessibility = std::min(1.0, (border ? 0.85 : 0.3) + noise(i));
97
```

```
98          double sp = 0.2;
99          for (auto& wh : waterholes) {
100             double d = std::hypot(z.row - wh.first, z.col - wh.second);
101             sp = std::max(sp, std::exp(-0.5 * d));
102         }
103         if (z.col < 4) sp = std::min(1.0, sp + 0.3);
104         z.species_risk = std::min(1.0, sp + noise(i*2));
105
106         z.wildlife_movement = std::min(1.0,
107             0.4 * z.species_risk + 0.3 * (1.0 - z.accessibility) + noise(i
    *3));
108
109         double pan = std::hypot(z.row - GRID_ROWS/2.0, z.col - GRID_COLS
    /2.0);
110         double max_pan = std::hypot(GRID_ROWS/2.0, GRID_COLS/2.0);
111         z.environmental = std::min(1.0, 0.5*(1.0 - pan/max_pan) + noise(i*4)
    );
112
113         z.h = W_SPECIES  * z.species_risk
114             + W_ACCESS   * z.accessibility
115             + W_WILDLIFE * z.wildlife_movement
116             + W_ENVIRON  * z.environmental;
117     }
118     return zones;
119 }
120
121 // Reachability matrix
122
123 std::vector<std::vector<int>> build_reach(const std::vector<Zone>& zones) {
124     std::vector<std::vector<int>> reach(N_ZONES);
125     for (int i = 0; i < N_ZONES; i++)
126         for (int j = 0; j < N_ZONES; j++) {
127             double d = std::hypot(zones[i].row - zones[j].row,
128                                   zones[i].col - zones[j].col);
129             if (d <= PATROL_RADIUS)
130                 reach[i].push_back(j);
131         }
132     return reach;
133 }
134
135 // Greedy patrol optimisation
136
137 double optimise(std::vector<Zone>& zones,
138                 const std::vector<std::vector<int>>& reach, int m) {
139     for (auto& z : zones) { z.covered = false; z.patrol_base = false; }
140     std::vector<bool> covered(N_ZONES, false);
141     double total = 0.0;
142
143     for (int step = 0; step < m; step++) {
144         double best = -1.0; int bi = -1;
145         for (int i = 0; i < N_ZONES; i++) {
146             if (zones[i].patrol_base) continue;
147             double gain = 0.0;
```

```
            for (int j : reach[i])
                if (!covered[j]) gain += zones[j].h;
            if (gain > best) { best = gain; bi = i; }
        }
        if (bi < 0) break;
        zones[bi].patrol_base = true;
        for (int j : reach[bi])
            if (!covered[j]) { covered[j] = true; total += zones[j].h; }
    }
    for (int j = 0; j < N_ZONES; j++) zones[j].covered = covered[j];
    return total;
}

// Wildfire Poisson model

double lambda_t(double t) {
    return LAMBDA_0 * (1.0 + SEASONAL_AMP *
           std::sin(2.0 * M_PI * t / 12.0 - PHASE_SHIFT));
}

double cumulative_lambda(double t1, double t2, int steps = 1000) {
    double dt = (t2-t1)/steps, sum = 0.0;
    for (int k = 0; k < steps; k++)
        sum += lambda_t(t1 + (k+0.5)*dt) * dt;
    return sum;
}

double fire_prob(double t1, double t2) {
    return 1.0 - std::exp(-cumulative_lambda(t1, t2));
}

// Wildlife exit model

double risk_index(double D, double Vn) {
    double Dr = 500.0;
    return ALPHA * std::exp(-DECAY * D) + BETA * std::max(Vn, 0.0) / Dr;
}

double alert_score(double RI) {
    return 1.0 / (1.0 + std::exp(-K_STEEP * (RI - THETA)));
}

double ttb(double D, double Vn) {
    return D / std::max(Vn, 0.01);
}

std::string alert_type(double D) {
    if (D <= 200)      return "CRITICAL";
    else if (D <= 500) return "CLOSE MONITORING";
    else               return "LIGHT MONITORING";
}

// Fire Risk Index
```

```
201
202 double compute_fri(const Zone& z, double weather) {
203     double pan = std::hypot(z.row - GRID_ROWS/2.0, z.col - GRID_COLS/2.0);
204     double max_pan = std::hypot(GRID_ROWS/2.0, GRID_COLS/2.0);
205     double f_fuel    = 1.0 - pan/max_pan;
206     double f_weather = weather;
207     double f_access  = z.accessibility;
208     return 0.40*f_fuel + 0.35*f_weather + 0.25*f_access;
209 }
210
211 // Main
212
213 int main() {
214     std::cout << std::fixed << std::setprecision(4);
215     std::vector<ModelResult> results;
216
217     // 1. Zone initialisation
218     auto zones = init_zones();
219     double max_h = 0.0;
220     for (auto& z : zones) max_h += z.h;
221
222     bool zones_ok = (zones.size() == N_ZONES && max_h > 0);
223     results.push_back({"Zone grid created (500 zones)", zones_ok,
224                        std::to_string(N_ZONES) + " zones"});
225
226     bool hj_ok = true;
227     for (auto& z : zones) if (z.h < 0 || z.h > 1) { hj_ok = false; break; }
228     results.push_back({"All h_j in [0,1]", hj_ok, "bounds satisfied"});
229
230     // 2. Reachability matrix
231     auto reach = build_reach(zones);
232     int avg_reach = 0;
233     for (auto& r : reach) avg_reach += r.size();
234     avg_reach /= N_ZONES;
235     results.push_back({"Reachability matrix built", true,
236                        "avg zones per patrol = " + std::to_string(avg_reach)
    });
237
238     // 3. Patrol optimisation
239     double wc = optimise(zones, reach, M_PATROL);
240     int covered = 0;
241     for (auto& z : zones) if (z.covered) covered++;
242     double frac = wc / max_h;
243     results.push_back({"Patrol model converged", true,
244                        std::to_string(covered) + "/" +
245                        std::to_string(N_ZONES) + " zones covered"});
246     results.push_back({"Weighted coverage >= 80%", frac >= 0.80,
247                        std::to_string(frac*100).substr(0,6) + "%"});
248     int bases = 0;
249     for (auto& z : zones) if (z.patrol_base) bases++;
250     results.push_back({"Sum x_i = 100 (resource constraint)",
251                        bases == M_PATROL,
252                        "deployed = " + std::to_string(bases)});
```

```
253
254     // 4. Wildfire model
255     double p_annual  = fire_prob(0.0, 12.0);
256     double p_sep_oct = fire_prob(8.0, 10.0);
257     results.push_back({"Annual fire probability computed", p_annual > 0,
258                        std::to_string(p_annual).substr(0,6)});
259     results.push_back({"Sep-Oct fire probability ~32-40%",
260                        p_sep_oct > 0.28 && p_sep_oct < 0.42,
261                        std::to_string(p_sep_oct*100).substr(0,5) + "%"});
262
263     // 5. Fire Risk Index
264     double mean_fri = 0; int high_risk = 0;
265     for (auto& z : zones) {
266         double fri = compute_fri(z, 0.85);
267         mean_fri += fri;
268         if (fri > 0.7) high_risk++;
269     }
270     mean_fri /= N_ZONES;
271     results.push_back({"FRI computed for all zones", true,
272                        "mean = " + std::to_string(mean_fri).substr(0,6)});
273     results.push_back({"High-risk zones (FRI > 0.7) identified",
274                        high_risk > 0,
275                        std::to_string(high_risk) + " zones"});
276
277     // 6. Wildlife exit model
278     struct Animal { std::string name; double D, Vn; };
279     std::vector<Animal> animals = {
280         {"Black Rhino A", 120.0, 3.5},
281         {"Elephant B",    380.0, 1.2},
282         {"Springbok C",   650.0, 0.3},
283         {"Lion D",         85.0, 5.8},
284         {"Oryx E",        490.0, 2.1},
285     };
286     bool exit_ok = true;
287     for (auto& a : animals) {
288         double RI = risk_index(a.D, a.Vn);
289         double AS = alert_score(RI);
290         if (RI < 0 || RI > 1) exit_ok = false;
291         if (AS < 0 || AS > 1) exit_ok = false;
292     }
293     results.push_back({"RI and AS in valid range [0,1]", exit_ok,
294                        "all 5 test animals verified"});
295     double as_lion = alert_score(risk_index(85.0, 5.8));
296     results.push_back({"Alert triggers for D<=200m, high Vn",
297                        as_lion > 0.45,
298                        "Lion D: AS = " + std::to_string(as_lion).substr(0,6)
299                        + " (alert issued)"});
300
301     // 7. Seasonal analysis
302     for (auto& z : zones) { z.covered = false; z.patrol_base = false; }
303     double dry_frac = optimise(zones, reach, M_PATROL) / max_h;
304     int wet_m = (int)(M_PATROL * 0.79);
305     for (auto& z : zones) { z.covered = false; z.patrol_base = false; }
```

```
    double wet_frac = optimise(zones, reach, wet_m) / max_h;
    results.push_back({"Dry season full coverage achieved",
                       dry_frac >= 0.99,
                       std::to_string(dry_frac*100).substr(0,6)+"%"});
    results.push_back({"Wet season coverage maintained > 85%",
                       wet_frac >= 0.85,
                       std::to_string(wet_frac*100).substr(0,6)+"%"});

    // Summary
    int passed = 0, failed = 0;
    for (auto& r : results) {
        print_result(r);
        if (r.passed) passed++; else failed++;
    }
    std::cout << "\nTotal: " << results.size()
              << "  Passed: " << passed
              << "  Failed: " << failed << "\n";
    if (failed == 0)
        std::cout << "All model components verified successfully.\n";

    return (failed == 0) ? 0 : 1;
}
```

Appendix B - Derivations

Derivations

1. Rothermel's Fire Spread Rate Derivation

Energy Balance Set Up:

1. Heat supplied from burning zone:

$$\dot{Q}_{in} = I_R \cdot \xi \cdot (1 + \phi_w + \phi_s)$$

2. Heat required to raise slab to ignition:

$$\dot{Q}_{req} = \rho_b \cdot \varepsilon \cdot Q_{ig} \cdot R$$

3. Reaction Intensity:

$$I_R = \Gamma' \cdot h \cdot \eta_M \cdot \eta_s$$

4. Propagating Flux Ratio:

$$\xi = \frac{\exp[(0.792 + 0.681\,\sigma^{0.5})(\beta + 0.1)]}{192 + 0.2595\,\sigma}$$

5. Wind Factor:

$$\phi_w = C \cdot U^B \cdot \left(\frac{\beta}{\beta_{op}}\right)^{-E}$$

6. Slope Factor:

$$\phi_s = 5.275\,\beta^{-0.3}\tan^2(\varphi)$$

2. Minimum Staffing

$$\max(C - \lambda S) \implies \frac{\partial C}{\partial S} = \lambda \implies C = \lambda S^*$$

$$S^* = \frac{\sum_{j \in Z} h_j y_j}{\lambda}$$

3. Poisson Distribution

Define mean arrivals over $[t_1, t_2]$:

$$\Theta(t_1, t_2) = \int_{t_1}^{t_2} \lambda(t)\,dt$$

Partition $[t_1, t_2]$ into n sub-intervals:

$$\Delta t = \frac{t_2 - t_1}{n}, \qquad p = \frac{\Theta(t_1, t_2)}{n}$$

Binomial probability:

$$P(N = k) = \binom{n}{k} \cdot \left(\frac{\Theta}{n}\right)^k \cdot \left(1 - \frac{\Theta}{n}\right)^{n-k}$$

Expand:

$$= \frac{n!}{k!(n-k)!} \cdot \frac{\Theta^k}{n^k} \cdot \left(1 - \frac{\Theta}{n}\right)^n \cdot \left(1 - \frac{\Theta}{n}\right)^{-k}$$

Take $n \to \infty$:

$$\frac{n(n-1)\cdots(n-k+1)}{n^k} \to 1$$

$$\left(1 - \frac{\Theta}{n}\right)^n \to e^{-\Theta}$$

$$\left(1 - \frac{\Theta}{n}\right)^{-k} \to 1$$

Result:

$$\boxed{P\big(N(t_1, t_2) = k\big) = \frac{\Theta(t_1, t_2)^k \cdot e^{-\Theta(t_1, t_2)}}{k!}}$$

4. Penalised Coverage Objective

Start with total weighted coverage:

$$\max \sum_{j \in Z} h_j y_j$$

Introduce personnel cost across both shifts:

$$\text{Total personnel} = x_D + x_N$$

Apply penalty coefficient λ on personnel use:

$$\text{Penalty} = \lambda(x_D + x_N)$$

Subtract penalty from weighted coverage:

$$\max \sum_{j \in Z} h_j y_j - \lambda(x_D + x_N)$$

Full objective function:

$$\boxed{\max \left(\sum_{j \in Z} h_j y_j - \lambda(x_D + x_N) \right)}$$